

PHASE 1 – PROJECT OVERVIEW

EduGenie – Google Gemini Powered Learning Assistant

Project Documentation | Page 1 of 8

Project Title

EduGenie – Google Gemini Powered Learning Assistant

Introduction

EduGenie is an AI-powered learning assistant designed to support students in understanding educational topics, asking questions, and receiving clear and useful explanations. The system uses Google Gemini to generate natural-language responses and provide interactive learning support.

Problem Statement

Students may find it difficult to get immediate explanations when learning independently. Traditional learning resources may not provide personalized responses to every question. EduGenie aims to provide a simple and interactive assistant that can respond to student queries.

Objectives

- Provide quick answers to learning-related questions.
- Explain concepts in a simple manner.
- Support interactive and personalized learning.
- Reduce the difficulty of finding relevant explanations.
- Demonstrate the practical use of Generative AI in education.

Project Scope

The project focuses on an educational assistant that accepts user questions and generates AI-based responses using Google Gemini.

PHASE 2 – REQUIREMENT ANALYSIS

EduGenie – Google Gemini Powered Learning Assistant

Project Documentation | Page 2 of 8

Functional Requirements

- User should be able to enter a question.
- System should process the question.
- System should communicate with the Gemini model/API as configured.
- System should display the generated response clearly.
- User should be able to ask multiple questions.

Non-Functional Requirements

- Simple and user-friendly interface.
- Reasonable response time.
- Reliable handling of normal user inputs.
- Maintainable and modular implementation.

Software Requirements

- Python / project programming environment as applicable
- Google Gemini / Generative AI service
- Required libraries and dependencies
- Web browser
- Code editor or development environment

Hardware Requirements

- Computer or laptop
- Internet connection
- Basic processor and memory suitable for development

Expected Outcome

A working learning assistant capable of accepting educational questions and presenting AI-generated explanations.

PHASE 3 – SYSTEM DESIGN

EduGenie – Google Gemini Powered Learning Assistant

Project Documentation | Page 3 of 8

System Architecture

The system follows a simple input-processing-output architecture. The user provides a question through the application interface. The application sends the request to the configured Gemini service. The generated response is then returned and displayed to the user.

Data Flow

User Question → Application Interface → Gemini Model/API → Generated Response → User Interface

Main Components

1. User Interface – accepts questions and displays answers.
2. Application Logic – validates and processes requests.
3. Gemini Integration – sends prompts to the AI model and receives responses.
4. Output Module – presents the response to the user.

Technologies Used

Google Gemini / Generative AI, programming language and libraries used in the implementation, development environment, and web/browser interface as applicable.

Architecture Evidence

SCREENSHOT PLACEHOLDER: Insert the system architecture / flow diagram here.

PHASE 4 – DATA / KNOWLEDGE PREPARATION

EduGenie – Google Gemini Powered Learning Assistant

Project Documentation | Page 4 of 8

Purpose

The assistant requires suitable prompts and learning-oriented content so that responses are relevant to educational use.

Input Preparation

Example learning inputs may include questions about subjects, definitions, explanations, examples, and short conceptual doubts.

Prompt Preparation

Prompts should clearly describe the student's question and request an educational response. The prompt can ask the model to explain a concept in simple language, provide examples, or present steps.

Sample Inputs

1. Explain a concept in simple words.
2. Give an example for a topic.
3. Explain the difference between two concepts.
4. Provide a short summary of a lesson.

Evidence

SCREENSHOT PLACEHOLDER: Insert sample questions/prompts or prepared learning content here.

PHASE 5 – IMPLEMENTATION

EduGenie – Google Gemini Powered Learning Assistant

Project Documentation | Page 5 of 8

Implementation Overview

The project implementation connects the application with the configured Google Gemini service. The application collects the user's question, prepares the request, sends it for generation, and displays the returned answer.

Implementation Steps

1. Create the project environment.
2. Install the required dependencies.
3. Configure the Gemini service/API as required.
4. Develop the user input interface.
5. Write the request and response handling logic.
6. Run and verify the application.

Important Modules

- User input handling
- Prompt/request preparation
- Gemini API/model integration
- Response handling
- Output display

Code Evidence

SCREENSHOT PLACEHOLDER: Insert the important project code screenshot here. Do not expose secret API keys in the screenshot.

Implementation Note

Keep API keys and other credentials private. Use environment variables or the secure configuration method recommended for the project.

PHASE 6 – TESTING

Testing Objective

Testing is performed to verify that the application accepts valid questions, communicates correctly with the AI service, and displays useful responses.

Test Cases

Test Case 1: Enter a normal educational question → Expected: relevant AI response → Result: Pass/Fail to be filled after testing.

Test Case 2: Enter another subject-related question → Expected: clear explanation → Result: Pass/Fail to be filled.

Test Case 3: Submit an empty input → Expected: validation/message if implemented → Result: Pass/Fail to be filled.

Testing Table

The following table can be completed using the actual results from your application.

Evidence

SCREENSHOT PLACEHOLDER: Insert testing screenshots and console/output evidence here.

PHASE 7 – RESULTS & EVIDENCE

EduGenie – Google Gemini Powered Learning Assistant

Project Documentation | Page 7 of 8

Result

The completed EduGenie prototype is intended to provide an interactive educational question-and-answer experience using Google Gemini. The final result should demonstrate that a user can enter a learning question and receive an AI-generated response.

Code Output

CODE OUTPUT SCREENSHOT PLACEHOLDER: Insert the terminal/console output or application output generated after running the code.

Application Output

APPLICATION SCREENSHOT PLACEHOLDER: Insert the final EduGenie interface showing a question and generated response.

Evidence Checklist

- Project running screenshot
- Code screenshot
- Code/console output screenshot
- Gemini response screenshot
- Final application screenshot

Observation

Testing evidence should be taken from the actual project execution. Replace all placeholders with your own screenshots before submission.

PHASE 8 – CONCLUSION & FUTURE SCOPE

EduGenie – Google Gemini Powered Learning Assistant

Project Documentation | Page 8 of 8

Conclusion

EduGenie demonstrates how Generative AI can be used to support students with interactive learning assistance. By integrating Google Gemini with an application interface, the project provides a practical approach for answering educational questions and explaining concepts.

Advantages

- Interactive learning support
- Quick AI-generated explanations
- Simple question-and-answer workflow
- Can support multiple learning topics
- Demonstrates practical Generative AI integration

Limitations

- AI responses may require verification.
- Response quality depends on the prompt and configured model/service.
- Internet/service availability may affect operation.

Future Scope

- Add subject-wise learning modes.
- Add quiz and practice-question generation.
- Add progress tracking.
- Add multilingual learning support.
- Add voice-based interaction.
- Improve personalization and learning analytics.

Final Statement

EduGenie provides a foundation for building an AI-assisted learning platform and can be extended with additional educational features in future versions.